

2.6 Moment generating functions and independence

Means and covariance matrices summarize a distribution. They do not determine it, and they cannot certify independence. For that we need a transform of the whole distribution. This section collects the facts that [Chapter 3](#) and [Chapter 4](#) use.

Definition 2.6.1 — Moment generating function

The **moment generating function** of a $p \times 1$ random vector $\mathbf{Y}$ is

$$M_{\mathbf{Y}}(\mathbf{t}) = \mathbb{E} \exp(\mathbf{t}^\top \mathbf{Y}), \quad \mathbf{t} \in \mathbb{R}^p,$$

where the expectation may be $+\infty$. We say that $M_{\mathbf{Y}}$ *exists* if it is finite for all $\mathbf{t}$ in some neighbourhood $\{\|\mathbf{t}\| < h\}$ of the origin, $h > 0$.

Theorem 2.6.1 — Properties of moment generating functions

Suppose $M_{\mathbf{Y}}$ exists.

- (a) *Uniqueness.* If $M_{\mathbf{V}}$ also exists and $M_{\mathbf{Y}} = M_{\mathbf{V}}$ on a neighbourhood of $\mathbf{0}$, then $\mathbf{Y}$ and $\mathbf{V}$ have the same distribution.
- (b) *Moments.* All moments of $\mathbf{Y}$ are finite. The gradient of $M_{\mathbf{Y}}$ at $\mathbf{0}$ is $\mathbb{E}(\mathbf{Y})$, and its Hessian at $\mathbf{0}$ is $\mathbb{E}(\mathbf{Y}\mathbf{Y}^\top)$. The Hessian of $\log M_{\mathbf{Y}}$ at $\mathbf{0}$ is $\text{Cov}(\mathbf{Y})$.
- (c) *Affine maps.* For a constant $k \times p$ matrix $\mathbf{A}$ and $k \times 1$ vector $\mathbf{b}$, $M_{\mathbf{A}\mathbf{Y}+\mathbf{b}}(\mathbf{s}) = \exp(\mathbf{s}^\top \mathbf{b}) M_{\mathbf{Y}}(\mathbf{A}^\top \mathbf{s})$.
- (d) *Independence.* Partition $\mathbf{Y} = (\mathbf{Y}_1^\top, \mathbf{Y}_2^\top)^\top$ and $\mathbf{t} = (\mathbf{t}_1^\top, \mathbf{t}_2^\top)^\top$ conformably. Then $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are independent iff $M_{\mathbf{Y}}(\mathbf{t}) = M_{\mathbf{Y}_1}(\mathbf{t}_1) M_{\mathbf{Y}_2}(\mathbf{t}_2)$ for all $\mathbf{t}$ in a neighbourhood of $\mathbf{0}$.

Proof. For (a), the scalar case is a theorem of analysis: a moment generating function that is finite near the origin extends analytically, which pins down the characteristic function and hence the distribution (Billingsley 1995). For vectors, fix $\mathbf{a}$. For small $|s|$, $M_{\mathbf{a}^\top \mathbf{Y}}(s) = M_{\mathbf{Y}}(\mathbf{s}\mathbf{a}) = M_{\mathbf{V}}(\mathbf{s}\mathbf{a}) = M_{\mathbf{a}^\top \mathbf{V}}(s)$, so $\mathbf{a}^\top \mathbf{Y}$ and $\mathbf{a}^\top \mathbf{V}$ have the same distribution by the scalar case, and [Proposition 2.6.2](#) below finishes the argument. For (b), finiteness near $\mathbf{0}$ implies $\mathbb{E} \exp(h'|Y_i|) < \infty$ for some $h' > 0$ and each i , so every polynomial in the components is integrable and differentiation under the expectation is justified. Then $\partial M / \partial t_i = \mathbb{E}(Y_i e^{\mathbf{t}^\top \mathbf{Y}})$ and $\partial^2 M / \partial t_i \partial t_j = \mathbb{E}(Y_i Y_j e^{\mathbf{t}^\top \mathbf{Y}})$. At $\mathbf{t} = \mathbf{0}$ these are $\mathbb{E}(Y_i)$ and $\mathbb{E}(Y_i Y_j)$. For $K = \log M$, the chain rule gives $\nabla^2 K = M^{-1} \nabla^2 M - M^{-2} (\nabla M)(\nabla M)^\top$, which at $\mathbf{0}$, where $M = 1$, is $\mathbb{E}(\mathbf{Y}\mathbf{Y}^\top) - \boldsymbol{\mu}\boldsymbol{\mu}^\top = \text{Cov}(\mathbf{Y})$. For (c), $\mathbb{E} \exp(\mathbf{s}^\top (\mathbf{A}\mathbf{Y} + \mathbf{b})) = e^{\mathbf{s}^\top \mathbf{b}} \mathbb{E} \exp((\mathbf{A}^\top \mathbf{s})^\top \mathbf{Y})$. For (d), independence gives $\mathbb{E}(e^{\mathbf{t}_1^\top \mathbf{Y}_1} e^{\mathbf{t}_2^\top \mathbf{Y}_2}) = \mathbb{E}(e^{\mathbf{t}_1^\top \mathbf{Y}_1}) \mathbb{E}(e^{\mathbf{t}_2^\top \mathbf{Y}_2})$. Conversely, let $\mathbf{V}_1$ and $\mathbf{V}_2$ be independent with the distributions of $\mathbf{Y}_1$ and $\mathbf{Y}_2$. The vector $(\mathbf{V}_1^\top, \mathbf{V}_2^\top)^\top$ has moment generating function $M_{\mathbf{Y}_1}(\mathbf{t}_1) M_{\mathbf{Y}_2}(\mathbf{t}_2)$, which by hypothesis equals $M_{\mathbf{Y}}$ near $\mathbf{0}$. By (a) it has the same joint distribution as $\mathbf{Y}$, so $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are independent. ■

The next fact explains why [Chapter 3](#) can define the multivariate normal through linear combinations.

Proposition 2.6.2 — Distributions are determined by linear combinations

If $\mathbf{a}^\top \mathbf{Y}$ and $\mathbf{a}^\top \mathbf{V}$ have the same distribution for every $\mathbf{a} \in \mathbb{R}^p$, then $\mathbf{Y}$ and $\mathbf{V}$ have the same distribution.

Proof. The characteristic function $\phi_{\mathbf{Y}}(\mathbf{t}) = \mathbb{E} \exp(i\mathbf{t}^\top \mathbf{Y})$ exists for every random vector and determines its distribution (Billingsley 1995). Since $\phi_{\mathbf{Y}}(\mathbf{t}) = \phi_{\mathbf{t}^\top \mathbf{Y}}(1) = \phi_{\mathbf{t}^\top \mathbf{V}}(1) = \phi_{\mathbf{V}}(\mathbf{t})$ for every $\mathbf{t}$, the two distributions coincide. ■

If $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are independent, so are $g(\mathbf{Y}_1)$ and $h(\mathbf{Y}_2)$ for measurable g and h . Independence of statistics is usually proved this way. For normal vectors, Theorem 3.4.1 reduces the independence of two blocks to a zero cross-covariance, and Theorem 4.4.1 and Theorem 4.4.2 build on that.

Independence of *several* vectors means that the joint distribution factors into all the marginals, not merely that each pair is independent.

Example. (Pairwise but not mutually independent)

Let S_1 and S_2 be independent, each equal to ± 1 with probability $\frac{1}{2}$, and put $S_3 = S_1 S_2$. Each S_j is ± 1 with probability $\frac{1}{2}$. Each pair is independent: for instance, $\Pr(S_1 = s, S_3 = u) = \Pr(S_1 = s, S_2 = su) = \frac{1}{4}$ for all signs s, u . So $\text{Cov}(\mathbf{S}) = \mathbf{I}_3$. But the three are not mutually independent, since S_3 is a function of the other two. A third-order moment shows the failure directly: $\mathbb{E}(S_1 S_2 S_3) = 1$, whereas mutual independence would force it to be 0. No condition on the covariance matrix can detect this, because covariances involve only pairs.

2.6.1 Exercises

2.6.1.1 A. Check your understanding

Exercise A1.

Find the moment generating function of the multinomial count vector of Example (Multinomial counts), and recover its mean and covariance matrix from [Theorem 2.6.1\(b\)](#).

2.6.1.2 B. Practice

Exercise B1.

Let $S_1, \dots, S_k$ be independent random signs. For each nonempty $A \subseteq \{1, \dots, k\}$ put $T_A = \prod_{i \in A} S_i$. Show that the $2^k - 1$ variables T_A are pairwise independent with covariance matrix $\mathbf{I}$, and find a triple of them that is not mutually independent.

2.6.1.3 C. Going deeper

Exercise C1.

Let φ be the $N(0, 1)$ density and, for $|a| \leq 1$, let

$$f_a(x) = \frac{\varphi(\log x)}{x} \{1 + a \sin(2\pi \log x)\}, \quad x > 0.$$

- (a) Show that f_a is a probability density for every such a , and that f_0 is the lognormal density, the density of e^W with $W \sim N(0, 1)$.
- (b) Show that a random variable X_a with density f_a has $E(X_a^k) = e^{k^2/2}$ for every integer $k \geq 0$, the same for all a . *Hint:* substitute $u = \log x$ and use $e^{ku}\varphi(u) = e^{k^2/2}\varphi(u - k)$, then the symmetry of φ .
- (c) Show that $E e^{tX_0} = \infty$ for every $t > 0$. Conclude that the uniqueness statement [Theorem 2.6.1\(a\)](#) cannot be weakened to “all moments are finite and agree”: these distributions differ, yet no moment, and hence no moment-based argument, can separate them.

Solution.

- (a) Since $|\sin| \leq 1$ and $|a| \leq 1$, the brace is nonnegative, so $f_a \geq 0$. The substitution $u = \log x$, $dx = x du$, turns $\int_0^\infty f_a(x) dx$ into $\int_{-\infty}^\infty \varphi(u) \{1 + a \sin(2\pi u)\} du = 1$, because φ is a density and $u \mapsto \varphi(u) \sin(2\pi u)$ is odd and integrable. The same substitution identifies f_0 as the density of e^W .
- (b) With the same substitution,

$$E(X_a^k) = \int_{-\infty}^\infty e^{ku} \varphi(u) \{1 + a \sin(2\pi u)\} du = e^{k^2/2} \left[1 + a \int_{-\infty}^\infty \varphi(u - k) \sin(2\pi u) du \right].$$

The remaining integral is $E \sin\{2\pi(W + k)\}$ with $W \sim N(0, 1)$, which expands as $\sin(2\pi k) E \cos(2\pi W) + \cos(2\pi k) E \sin(2\pi W)$. For integer k the first coefficient $\sin(2\pi k)$ is zero, and $E \sin(2\pi W) = 0$ because W is symmetric. So $E(X_a^k) = e^{k^2/2}$ whatever a is.

- (c) For $t > 0$, $E e^{tX_0} = \int_{-\infty}^\infty \exp(te^u) \varphi(u) du$, and $te^u - u^2/2 \rightarrow \infty$ as $u \rightarrow \infty$, so the integrand does not even tend to zero: the integral diverges. Hence [Theorem 2.6.1](#) does not apply to the lognormal law, and indeed it cannot: the perturbed laws have exactly the same moments of every order but different densities, so they are genuinely different distributions.

Exercise C2.

- (a) Suppose each component of $\mathbf{Y}$ has a moment generating function that is finite on $|s| < h$. Show that $M_{\mathbf{Y}}(\mathbf{t})$ is finite whenever $\max_i |t_i| < h/p$. *Hint:* convexity of the exponential gives $\exp(\sum_i t_i Y_i) \leq p^{-1} \sum_i \exp(pt_i Y_i)$. So the hypothesis of [Theorem 2.6.1](#) is a statement about the components, but its conclusions are not.
- (b) Let $Z \sim N(0, 1)$, let S be an independent random sign, and put $W = SZ$. Show that

$$M_{(Z,W)}(\mathbf{t}) = \frac{1}{2} \exp\left\{\frac{1}{2}(t_1 + t_2)^2\right\} + \frac{1}{2} \exp\left\{\frac{1}{2}(t_1 - t_2)^2\right\}.$$

Deduce that Z and W are each $N(0, 1)$, that $\text{Cov}(Z, W) = 0$, and that they are

not independent.

- (c) Compute $E(Z^2 W^2)$ and compare it with the value it would have under independence.

Solution.

- (a) Each pt_i lies in $(-h, h)$, so $E \exp(pt_i Y_i) = M_{Y_i}(pt_i) < \infty$. Averaging the hint over i and taking expectations gives $M_{\mathbf{Y}}(\mathbf{t}) \leq p^{-1} \sum_i M_{Y_i}(pt_i) < \infty$. The hint itself is the convexity inequality $\exp(p^{-1} \sum_i s_i) \leq p^{-1} \sum_i e^{s_i}$ with $s_i = pt_i Y_i$.
- (b) Conditioning on S , $E \exp(t_1 Z + t_2 SZ) = \frac{1}{2} E \exp\{(t_1 + t_2)Z\} + \frac{1}{2} E \exp\{(t_1 - t_2)Z\}$, and each term is a normal moment generating function, which gives the display. Setting $t_2 = 0$ leaves $e^{t_1^2/2}$, so $Z \sim N(0, 1)$; setting $t_1 = 0$ leaves the same, so $W \sim N(0, 1)$. Expanding to second order, $M_{(Z, W)}(\mathbf{t}) = 1 + \frac{1}{2}(t_1^2 + t_2^2) + O(\|\mathbf{t}\|^4)$ with no $t_1 t_2$ term, so $\text{Cov}(Z, W) = 0$ by [Theorem 2.6.1\(b\)](#). Were they independent, the joint moment generating function would be $\exp\{\frac{1}{2}(t_1^2 + t_2^2)\}$ by [Theorem 2.6.1\(d\)](#). Along $\mathbf{t} = (s, s)^\top$ the display equals $\frac{1}{2}(e^{2s^2} + 1) = 1 + s^2 + s^4 + O(s^6)$, whereas $\exp(s^2) = 1 + s^2 + \frac{1}{2}s^4 + O(s^6)$. The two disagree in every neighbourhood of the origin, so Z and W are dependent, although both are normal and uncorrelated.
- (c) $W^2 = Z^2$, so $E(Z^2 W^2) = E(Z^4) = 3$, whereas independence would give $E(Z^2) E(W^2) = 1$. This is the failure of Example (A point on a circle) in a normal disguise; joint normality, defined in [Chapter 3](#), is exactly what rules it out.